HU Collection

TMUA Mock Set A, Paper 2

Written by Kevin Hu

Recommended time:	2 hours (the live TMUA allows 75 minutes)
Calculators:	not permitted
Format:	20 multiple-choice questions

This is an original mock paper written in the style of Paper 2 of the Test of Mathematics for University Admission (TMUA). Paper 2 tests mathematical reasoning: the logic of statements and their negations, necessary and sufficient conditions, counterexamples, the structure of proofs, and the ability to spot exactly where an argument goes wrong. The mathematical content stays within the Paper 1 syllabus, but the questions reward care with quantifiers, implications and edge cases rather than speed of calculation.

Guidance

The questions here sit at the hard end of the TMUA range. Several contain a detail that is easy to skim past, an isolated case, a quantifier in the wrong place, an assumption that looks harmless. Reading a question twice is time well spent on this paper.

The following conventions are used throughout. "A only if B" means "if A then B". "A unless B" means "if B does not hold, then A holds". A counterexample to a statement of the form "if A then B" is a case in which A holds and B fails. In a question that asks for the first mistake in an argument, a line counts as a mistake if it asserts something false or draws a conclusion that does not follow from the lines before it.

Each question has exactly one correct option. No marks are lost for a wrong answer, so every question should be attempted. Answers and full worked solutions are published separately on the HU Collection website. If you spot an error, or find a neater method, I would be glad to hear from you.

Good luck.

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Question 1

Consider the statement

For every real number x, there exists a real number y such that if $x < y$ then $x^2 < y^2$.

Which of the following is the negation of this statement?

A There exists a real number x such that for every real number y, if $x < y$ then $x^2 \geq y^2$.

B For every real number x, there exists a real number y such that $x < y$ and $x^2 \geq y^2$.

C There exists a real number x such that for every real number y, $x < y$ and $x^2 \geq y^2$.

D There exist real numbers x and y such that $x < y$ and $x^2 \geq y^2$.

E There exists a real number x such that for every real number y, $x \geq y$ or $x^2 \geq y^2$.

F For every real number x, there exists a real number y such that $x \geq y$ or $x^2 \geq y^2$.

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Question 2

Consider the following statement about real numbers x and y.

If $x+y$ and xy are both integers, then x and y are both integers.

Which of the following are counterexamples to this statement?

I $x=\sqrt{2}$ and $y=-\sqrt{2}$

II $x=\frac{1}{2}$ and $y=\frac{1}{2}$

III $x=\frac{3}{2}$ and $y=\frac{1}{2}$

IV $x=2+\sqrt{3}$ and $y=2-\sqrt{3}$

A none of them

B I only

C IV only

D I and IV only

E I, III and IV only

F II and III only

G I, II and IV only

H I, II, III and IV

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Question 3

A door is fitted with three switches, coloured red, blue and green, each of which is either on or off. Whether the door is locked or unlocked depends only on the positions of the three switches. It is known that:

If the green switch is on, or the red switch is on and the blue switch is off, then the door is unlocked.

Which one of the following statements must be true?

A If the door is unlocked, then the green switch is on, or the red switch is on and the blue switch is off.

B If the door is locked, then the green switch is off, the red switch is off and the blue switch is on.

C If the door is locked, then the green switch is off and the red switch is off.

D If the green switch is off and the blue switch is on, then the door is locked.

E If the door is locked, then the green switch is off, and either the red switch is off or the blue switch is on.

F If the door is unlocked and the green switch is off, then the red switch is on and the blue switch is off.

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Question 4

Let S be the statement

$$x > 2 \text{ only if } x^2 > 4,$$

where x is a real number. Which of the following statements are logically equivalent to S?

I $x^2 > 4$ is a necessary condition for $x > 2$.

II $x > 2$ is a necessary condition for $x^2 > 4$.

III $x^2 \leq 4$ unless $x > 2$.

IV $x \leq 2$ unless $x^2 > 4$.

A I only

B I and IV only

C IV only

D I and II only

E II and III only

F I, III and IV only

G III and IV only

H I, II, III and IV

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Question 5

Let b and c be real numbers, and let P be the statement

the equation $x^2 + bx + c = 0$ has two distinct positive real roots.

Which one of the following is a sufficient but not necessary condition for P?

A $b < 0$ and $c > 0$

B $b^2 > 4c$ and $c > 0$

C $b < 0$ and $b^2 > 4c$

D $c \geq 0$ and $b + 2\sqrt{c} < 0$

E $b < 0$, $c > 0$ and $b^2 > 4c$

F $b \leq -4$ and $0 < c \leq 3$

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Question 6

In a group of 40 students, 30 study French, 32 study German and 34 study Spanish. What is the smallest possible number of students who study all three languages?

A 6

B 10

C 12

D 14

E 16

F 18

G 20

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Question 7

Let a, b and c be positive real numbers, and consider the statements

$$P\text{: there is a triangle whose side lengths are } a, b \text{ and } c;$$
$$Q\text{: } a^2 + b^2 + c^2 < 2(ab + bc + ca).$$

Which of the following best describes the relationship between P and Q?

A P is necessary and sufficient for Q.

B P is necessary but not sufficient for Q.

C P is sufficient but not necessary for Q.

D P is neither necessary nor sufficient for Q.

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Question 8

The function f is defined for all real x by

$$f(x) = |x - 1| + |x + 1| - 2|x|.$$

Which of the following statements are true?

I $f(-x) = f(x)$ for all real x.

II The equation $f(x) = 1$ has exactly two real solutions.

III The equation $f(x) = 0$ has exactly two real solutions.

IV $f(x) \leq 2$ for all real x, with equality for exactly one value of x.

A I only

B I and II only

C I and IV only

D II and III only

E I, III and IV only

F II, III and IV only

G I, II, III and IV

H I, II and IV only

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Question 9

How many of the following six statements about real numbers are true?

(i) For every real x there exists a real y such that $x^2 + y^2 = 1$.

(ii) There exists a real y such that for every real x, $xy = x$.

(iii) For every real x there exists a real y such that $x + y^3 = 0$.

(iv) There exists a real x such that for every real y, $x \leq y^2$.

(v) For every real y there exists a real x such that $x^2 < y$.

(vi) There exists a real x such that for every real y, $x^2 < y^2 + 1$.

A 1

B 2

C 3

D 4

E 5

F 6

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Question 10

The following three facts are known about the students of a school.

Every member of the chess club is in Year 12 or in Year 13.
No Year 12 student studies Art.
Some members of the chess club study Art.

Which of the following statements can be deduced from these facts?

I Some Year 13 students study Art.

II No Year 12 student is a member of the chess club.

III Some members of the chess club are in Year 13.

IV Every member of the chess club who studies Art is in Year 13.

A I only

B I, III and IV only

C I and III only

D III and IV only

E I, II and III only

F II, III and IV only

G I, II, III and IV

H none of them

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Question 11

The real numbers a, b and c satisfy $a > b > c > 0$. Which of the following statements are necessarily true?

I $a^2 - b^2 > b^2 - c^2$

II $\frac{a}{b} + \frac{b}{c} > 2$

III $(a-b)(b-c) \leq \left(\frac{a-c}{2}\right)^2$

IV $\sqrt{a} - \sqrt{b} > \sqrt{b} - \sqrt{c}$

A none of them

B II and III only

C II only

D III only

E I and III only

F I, II and III only

G II, III and IV only

H I, II, III and IV

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Question 12

The polynomial f satisfies $\int_0^1 f(x)\,dx = 0$. Which of the following statements must be true?

I $f(c) = 0$ for some c with $0 \leq c \leq 1$.

II $\int_0^1 \big(f(x)\big)^2\,dx = 0$.

III $\int_0^1 x\,f(x)\,dx = 0$.

IV $\int_0^1 f(1-x)\,dx = 0$.

A I and IV only

B I only

C IV only

D I and II only

E I, III and IV only

F II and III only

G I, II, III and IV

H none of them

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Question 13

Let n be a positive integer, and consider the statements

P: $n^2 + 1$ is divisible by 5; Q: $n^4 - 1$ is divisible by 5; R: $n^2 - 1$ is divisible by 5.

Which one of the following is correct?

A Q holds if and only if at least one of P and R holds.

B P is a necessary condition for Q.

C R is a necessary condition for Q.

D Q is a sufficient condition for P.

E Q is a necessary condition for P, and R is a necessary condition for Q.

F Neither P nor R is a sufficient condition for Q.

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Question 14

A student attempts to prove the following statement.

If a sequence of real numbers $u_1, u_2, u_3, \ldots$ satisfies $u_{n+1} > u_n$ for every positive integer n, then for every real number M there is a term of the sequence that is greater than M.

The attempted proof is given line by line.

(1) Suppose that $u_{n+1} > u_n$ for every positive integer n.

(2) For each n, let $d_n = u_{n+1} - u_n$, so that $d_n > 0$.

(3) Let d be the smallest of the numbers $d_1, d_2, d_3, \ldots$; then $d > 0$.

(4) Then $u_n = u_1 + d_1 + d_2 + \cdots + d_{n-1} \geq u_1 + (n-1)d$ for every $n \geq 2$.

(5) Given any real number M, choose an integer n with $n > 1 + \dfrac{M - u_1}{d}$; then $u_1 + (n-1)d > M$.

(6) Hence, for this n, $u_n > M$.

(7) Since M was arbitrary, the terms of the sequence exceed any given real number.

Which one of the following is correct?

A The first mistake is in line (2), and the statement is false.

B The first mistake is in line (3), and the statement is true.

C The first mistake is in line (3), and the statement is false.

D The first mistake is in line (4), and the statement is false.

E The first mistake is in line (5), and the statement is true.

F The first mistake is in line (5), and the statement is false.

G The first mistake is in line (6), and the statement is false.

H The proof is valid, and the statement is true.

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Question 15

Seven distinct integers are chosen. Which of the following statements must be true?

I Two of the integers leave the same remainder on division by 6.

II The difference between some two of the integers is a multiple of 6.

III The sum of some two of the integers is a multiple of 6.

IV The sum of some three of the integers is a multiple of 3.

A I, II and IV only

B I and II only

C I, II and III only

D II and IV only

E IV only

F I, II, III and IV

G I and IV only

H none of them

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Question 16

A sequence is defined by $u_1 = 1$ and

$$u_{n+1} = u_n + \frac{1}{u_n} \qquad \text{for } n \geq 1.$$

Which of the following statements are true?

I $u_{n+1} > u_n$ for every positive integer n.

II $u_n^2 \geq 2n$ for every positive integer n.

III $u_n \leq n$ for every positive integer n.

IV $u_n > 100$ for some positive integer n.

A I only

B I and III only

C I and IV only

D I, II and III only

E I, II and IV only

F I, III and IV only

G I, II, III and IV

H II and IV only

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Question 17

The polynomial f has the property that $f'(x) > 0$ for every real x except $x = 2$, where $f'(2) = 0$. Which of the following statements must be true?

I $f(x) < f(y)$ whenever $x < y$.

II $f''(2) > 0$.

III The equation $f(x) = 0$ has exactly one real solution.

IV f has odd degree.

A I only

B I and III only

C I and IV only

D I, II and III only

E I, II and IV only

F I, II, III and IV

G I, III and IV only

H none of them

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Question 18

A student attempts to prove the following statement.

If p is a polynomial such that $p(n) = 2^n$ for each integer n with $0 \leq n \leq 10$, then $p(11) = 2^{11}$.

The attempted proof is given line by line.

(1) Define $q(x) = p(x) - 2^x$ for all real x.

(2) Then $q(n) = 0$ for $n = 0, 1, 2, \ldots, 10$, so q has at least 11 distinct roots.

(3) A non-zero polynomial with at least 11 distinct roots has degree at least 11.

(4) The polynomial p is determined by its values at the 11 points $0, 1, \ldots, 10$, so p has degree at most 10, and hence so does q.

(5) By (3) and (4), q must be the zero polynomial.

(6) So $p(x) = 2^x$ for all real x, and in particular $p(11) = 2^{11}$.

Which one of the following is correct?

A The first mistake is in line (1), and the statement is false.

B The first mistake is in line (2), and the statement is false.

C The first mistake is in line (3), and the statement is false.

D The first mistake is in line (3), and the statement is true.

E The first mistake is in line (4), and the statement is false.

F The first mistake is in line (4), and the statement is true.

G The first mistake is in line (5), and the statement is false.

H The proof is valid, and the statement is true.

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Question 19

Five statements S_1, S_2, S_3, S_4, S_5 are written on a board. For each $k = 1, 2, 3, 4, 5$, the statement S_k reads:

"At least k of the statements S_1, S_2, S_3, S_4, S_5 are false."

Which one of the following is correct?

A Exactly one of the statements is true.

B Exactly two of the statements are true.

C Exactly three of the statements are true.

D Exactly four of the statements are true.

E All five statements are false.

F All five statements are true.

G There is no way to assign truth values to the five statements consistently.

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Question 20

Let f and g be functions defined for all real numbers and taking real values, and consider the statements

$$P\text{: } f(g(x)) = x \text{ for every real } x; \qquad Q\text{: } g(f(x)) = x \text{ for every real } x.$$

Which one of the following is correct?

A P implies Q, but Q does not imply P.

B Q implies P, but P does not imply Q.

C P and Q are equivalent.

D Neither of P and Q implies the other.

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
	C	D	E	B	F	E	C	H	D	B

Q	11	12	13	14	15	16	17	18	19	20
	B	A	A	C	A	F	G	E	G	D

Worked solutions

Question 1

The statement has the shape "for every x there exists y such that (if A then B)". Negating it swaps each quantifier and negates the inside: "there exists x such that for every y, not (if A then B)". An implication fails exactly when its hypothesis holds and its conclusion fails, so "not (if A then B)" is "A and not B". The negation is therefore: there exists x such that for every y, $x < y$ and $x^2 \geq y^2$. As a sanity check, the original statement is true (given x, take $y = x - 1$, so the hypothesis $x < y$ fails and the implication holds), and the negation is visibly false, since no x satisfies $x < y$ for every y. Option A keeps the implication, B keeps the wrong quantifier order, D has two existentials, and E, F negate a different statement. **Answer C.**

Question 2

A counterexample must satisfy the hypothesis and fail the conclusion. In I, $x+y = 0$ and $xy = -2$ are integers while neither x nor y is: a counterexample. In II the product is $\frac{1}{4}$, and in III it is $\frac{3}{4}$, so in both the hypothesis fails and nothing is tested. In IV, $x + y = 4$ and $xy = 4 - 3 = 1$, both integers, while x and y are irrational: a counterexample. **Answer D.**

Question 3

Write R, B, G for "the red, blue, green switch is on" and U for "the door is unlocked". The given fact is $\big(G$ or $(R$ and not $B)\big) \Rightarrow U$. The only thing that must follow is its contrapositive, not $U \Rightarrow$ not$\big(G$ or $(R$ and not $B)\big)$, and by De Morgan the right-hand side is (not G) and $\big($(not R) or $B\big)$. That is option E. Option A is the converse. Options B and C draw conclusions stronger than the contrapositive allows: with the red and blue switches both on and the green off, the hypothesis is false and the door may be locked, which contradicts C. Options D and F are converse-type statements and need not hold. **Answer E.**

Question 4

"A only if B" means $A \Rightarrow B$, so S is $x > 2 \Rightarrow x^2 > 4$. Statement I says $x^2 > 4$ is necessary for $x > 2$, which is the same implication. Statement II reverses it, and is false ($x = -3$). With the stated reading of "unless", III is "if $x \leq 2$ then $x^2 \leq 4$", the inverse of S, and false ($x = -3$ again). Statement IV is "if $x^2 \leq 4$ then $x \leq 2$", the contrapositive of S, and equivalent to it. **Answer B.**

Question 5

The equation has two distinct positive real roots exactly when the discriminant is positive, the sum of the roots $-b$ is positive and the product c is positive: that is, $b < 0$, $c > 0$ and $b^2 > 4c$. Option E is this condition, so it is necessary and sufficient. Options A, B and C each drop one of the three requirements, so each is necessary but not sufficient (for A take $b = -1$, $c = 1$; for B take $b = 5$, $c = 1$; for C take $b = -1$, $c = -1$). Option D looks equivalent but is not: with $c = 0$ and $b = -1$ it holds, yet the roots are 0 and 1, so it is necessary but not sufficient. Option F forces $b < 0$,

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$c > 0$ and $b^2 \geq 16 > 12 \geq 4c$, so it is sufficient, and $b = -5$, $c = 6$ (roots 2 and 3) shows it is not necessary. **Answer F.**

Question 6

Count the students who miss each language: 10 do not study French, 8 do not study German and 6 do not study Spanish. A student who fails to study all three belongs to at least one of these groups, so at most $10 + 8 + 6 = 24$ students miss a language, and at least $40 - 24 = 16$ study all three. The bound is attained when the three groups are disjoint. **Answer E.**

Question 7

If a, b, c are the sides of a triangle then $a < b + c$, so $a^2 < ab + ac$, and similarly $b^2 < ab + bc$ and $c^2 < ac + bc$. Adding gives Q, so P is sufficient for Q. It is not necessary: $a = b = 1$, $c = 3$ gives $1 + 1 + 9 = 11 < 14 = 2(1 + 3 + 3)$, but $1 + 1 < 3$ so no triangle exists. **Answer C.**

Question 8

For $x \geq 1$ all three moduli open positively and $f(x) = 0$; the same holds for $x \leq -1$. For $0 \leq x \leq 1$, $f(x) = (1 - x) + (x + 1) - 2x = 2 - 2x$, and by symmetry $f(x) = 2 + 2x$ on $-1 \leq x \leq 0$. So the graph is a tent of height 2 on $[-1, 1]$ and zero outside. Statement I holds since $f(-x)$ is the same expression with the first two terms swapped. Statement II: $2 - 2|x| = 1$ gives $x = \pm\frac{1}{2}$ only. Statement III fails, since $f(x) = 0$ for every x with $|x| \geq 1$. Statement IV holds, with equality only at $x = 0$. **Answer H.**

Question 9

(i) is false: for $x = 2$ we would need $y^2 = -3$. (ii) is true with $y = 1$. (iii) is true with $y = -x^{1/3}$. (iv) is true with $x = 0$, since $0 \leq y^2$ for all y. (v) is false: for $y = 0$ there is no x with $x^2 < 0$. (vi) is true with $x = 0$. Four statements are true. **Answer D.**

Question 10

By the third fact there is a chess club member who studies Art. By the second fact this student is not in Year 12, so by the first fact they are in Year 13. This gives I and III at once, and the same argument applied to an arbitrary Art-studying member gives IV. Statement II does not follow: the facts say nothing against a Year 12 student who does not study Art being in the club. **Answer B.**

Question 11

Statement II holds because $a/b > 1$ and $b/c > 1$. Statement III is AM-GM applied to the positive numbers $a - b$ and $b - c$, whose sum is $a - c$. Statement I fails for $a = 4$, $b = 3$, $c = 1$, where the two sides are 7 and 8. Statement IV fails for $a = 1.21$, $b = 1$, $c = 0.01$, where the two sides are 0.1 and 0.9. **Answer B.**

Question 12

If f had no zero on $[0, 1]$ then, being a polynomial, it would keep one sign throughout the interval and its integral could not be zero, so I holds. Statement IV holds by the substitution $u = 1 - x$, which maps $[0, 1]$ onto itself and reverses orientation. Statements II and III both fail for $f(x) = 2x - 1$, which integrates to zero on $[0, 1]$ while $\int_0^1 (2x - 1)^2 \, dx = \frac{1}{3}$ and $\int_0^1 x(2x - 1) \, dx = \frac{1}{6}$. **Answer A.**

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Question 13

Since $n^4 - 1 = (n^2 - 1)(n^2 + 1)$ and 5 is prime, 5 divides $n^4 - 1$ exactly when it divides one of the factors, so Q holds if and only if P or R holds: option A is correct. To see that the others fail, note that squares leave remainder 0, 1 or 4 on division by 5, so P means $n \equiv \pm 2$, R means $n \equiv \pm 1$ and Q means $n \not\equiv 0$. Then $n = 1$ satisfies Q but not P (against B and D), $n = 2$ satisfies Q but not R (against C and the second half of E), and P alone is sufficient for Q (against F). **Answer A.**

Question 14

The statement is false: $u_n = 1 - \frac{1}{n}$ is increasing but every term is below 1. Lines (1) and (2) are correct. Line (3) is the first mistake: an infinite collection of positive numbers need not have a smallest member. For the sequence above the differences are $\frac{1}{n(n+1)}$, which get arbitrarily small without ever reaching a minimum. Everything after line (3) is correct reasoning from a quantity d that need not exist. **Answer C.**

Question 15

Statements I and II say the same thing, since two integers differ by a multiple of 6 exactly when they leave the same remainder, and both follow from the pigeonhole principle: seven integers, six remainders. Statement III fails: $1, 2, 7, 8, 13, 14, 19$ all leave remainder 1 or 2, so any two of them sum to remainder 2, 3 or 4. Statement IV holds: consider remainders on division by 3. If some remainder occurs three times, those three integers sum to a multiple of 3. Otherwise each of $0, 1, 2$ occurs at most twice, so with seven integers all three remainders occur, and one integer from each class sums to $0 + 1 + 2 \equiv 0$. **Answer A.**

Question 16

Every term is positive, so $u_{n+1} - u_n = 1/u_n > 0$ and I holds. Statement II fails at the boundary: $u_1^2 = 1 < 2$. Statement III holds by induction, since $u_1 = 1$ and if $u_n \le n$ then $u_{n+1} = u_n + 1/u_n \le n + 1$, using $u_n \ge u_1 = 1$. For IV, square the recurrence: $u_{n+1}^2 = u_n^2 + 2 + 1/u_n^2 > u_n^2 + 2$, so $u_n^2 > 2n - 1$ for $n \ge 2$, and u_n exceeds 100 as soon as $2n - 1 > 10000$. **Answer F.**

Question 17

Since $f' \ge 0$ everywhere and vanishes at a single point, f is strictly increasing, giving I. For II, f' has a minimum at $x = 2$, so $f''(2) = 0$, and $f''(2) > 0$ is impossible. For IV, f' is a non-constant polynomial that is positive for all large $|x|$ in both directions, so it has even degree, and f has odd degree. For III, a strictly increasing function has at most one root, and a polynomial of odd degree has at least one. **Answer G.**

Question 18

Lines (1) and (2) are correct: q is a function with eleven zeros. Line (3) is a true fact about polynomials. Line (4) is the first mistake. Knowing p at eleven points does not bound its degree: the interpolating polynomial of degree at most 10 exists, but so does that polynomial plus any multiple of $x(x-1)(x-2)\cdots(x-10)$. The line also treats q as a polynomial, which it is not, and line (5) compounds this by applying (3) to q. The statement itself is false. The polynomial $p(x) = \sum_{k=0}^{10} \binom{x}{k}$, where $\binom{x}{k} = \frac{x(x-1)\cdots(x-k+1)}{k!}$, has degree 10 and satisfies $p(n) = \sum_{k=0}^{10} \binom{n}{k} = 2^n$ for $0 \le n \le 10$, while $p(11) = 2^{11} - \binom{11}{11} = 2047$. **Answer E.**

Question 19

Suppose exactly f of the five statements are false. Then S_k is true precisely when $f \ge k$, so the true statements are $S_1, \ldots, S_f$ and the false ones are $S_{f+1}, \ldots, S_5$. Hence exactly $5 - f$ statements are

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false. Consistency requires $5 - f = f$, which no integer satisfies. So no assignment of truth values is consistent. (The familiar version of this puzzle uses "exactly k", which does have a consistent solution; the change to "at least" is what breaks it.) **Answer G.**

Question 20

Neither implication holds. Let $g(x) = 2^x$, and let $f(x) = \log_2 x$ for $x > 0$ with $f(x) = 0$ for $x \leq 0$. Then $f(g(x)) = \log_2 2^x = x$ for every x, so P holds, but $g(f(-1)) = g(0) = 1 \neq -1$, so Q fails. Swapping the roles of f and g in this example gives Q without P. The underlying point is that P says g is injective and f is surjective, while Q says the reverse; only when both hold is one function the inverse of the other. **Answer D.**

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